INTERNSHIP – PROJECT PHASE REPORT

Project chosen: HR Analytics - Predict Employee Attrition

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INTRODUCTION:

Employee attrition is a critical challenge for organizations, driving up recruitment and training costs while disrupting team continuity. This project analyzes IBM's HR Analytics dataset (1,470 employees) to identify which employees are likely to resign and understand the key factors driving attrition, enabling proactive retention strategies.

ABSTRACT:

This project applies exploratory data analysis, classification modeling, and SHAP explainability to predict employee attrition. A Logistic Regression model (class_weight='balanced') was selected over Random Forest for its superior Recall (72.3%) on identifying actual churners. SHAP analysis revealed OverTime, frequent Business Travel, and low tenure as the top attrition drivers. An interactive Power BI dashboard visualizes attrition patterns across departments, salary bands, and promotion timing, alongside a high-risk employee watchlist for targeted retention.

TOOLS USED:

- Python (Pandas, Seaborn, Matplotlib, Scikit-learn, SHAP)
- Power BI
- Jupyter Notebook

STEPS INVOLVED

1. EDA: analyzed attrition patterns by department, role, overtime, salary band, and promotion timing; addressed class imbalance (84% No / 16% Yes)
2. Preprocessing: one-hot encoding, 80/20 stratified split, class-weighted balancing
3. Modeling: compared Logistic Regression vs Random Forest; selected LR based on Recall/F1 (not accuracy, given imbalance)
4. Explainability: SHAP summary and force plots to identify global and individual attrition drivers
5. Power BI dashboard: KPIs, attrition breakdowns, high-risk employee table, slicers

KEY EDA FINDINGS:

1. OverTime employees churn at ~30.5% vs 10.4% for those who don't — the strongest single driver.
2. Salary Band: Low-income employees churn at 29.3% vs 10.3% for Very High income.
3. Promotion Trend: U-shaped pattern — 18.9% attrition at 0 years since promotion, dropping to 10.1% at 3-5 years, rising again to 16.3% at 5+ years.
4. JobRole: Sales Representatives churn highest (~40%), far above Managers and Research Directors (~3-5%).
5. Department: Sales (20.6%) and HR (19.1%) churn well above R&D (13.8%).
6. MaritalStatus: Single employees churn at 25.5% vs 12.4% (Married) and 10.1% (Divorced).
7. JobSatisfaction: Attrition decreases roughly monotonically from 22.8% (level 1) to 11.3% (level 4)

CONCLUSION:

Model Performance: Logistic Regression (class_weight='balanced') was selected as the final model over Random Forest, achieving Recall of 72.3%, F1-score of 0.523, ROC-AUC of 0.821, and Accuracy of 78.9%. The model correctly identified 34 of 47 actual churners in the test set.
Top 3 Attrition Drivers (SHAP):

1. OverTime = Yes
2. Frequent Business Travel
3. Low Total Working Years (early-career employees)

HR Recommendations:

1. Cap or closely monitor overtime for high-risk roles (Sales Representatives, Sales Executives) — OverTime=Yes employees churn at nearly 3x the rate of those without (30.5% vs 10.4%).
2. Reduce frequent business travel requirements, or offer compensating support (travel stipends, flexible scheduling), for roles requiring frequent travel, given its strong SHAP impact on attrition risk.
3. Strengthen early-career retention programs (mentorship, structured check-ins, career-path clarity) for low-tenure employees, since this is a top global driver and aligns with the "0 years since last promotion" spike (18.9% attrition) in the promotion-timing analysis.